

Inferential Statistics

Part-1

Sourav Karmakar

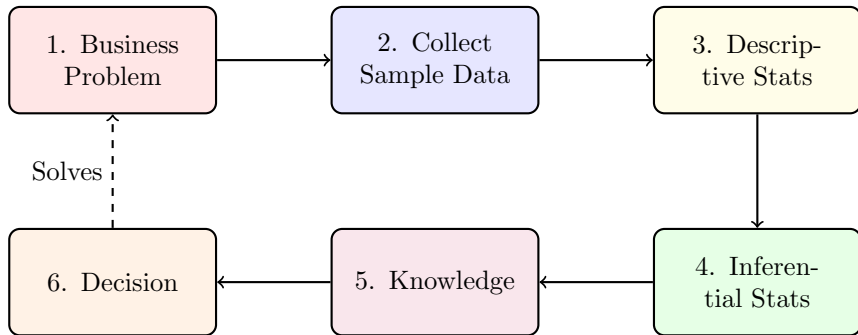
Outline

- 1 Basics of Inferential Statistics
- 2 Estimation Theory
- 3 Sampling Strategies
- 4 Sampling Distribution & CLT
- 5 Quiz

Part 1: The Statistical Framework

The Statistical Decision Making Process

In business and science, we rarely have all the data. We must make decisions based on partial information.



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- ❺ **Knowledge:** We are 95% confident Red is better.
- ❻ **Decision:** Roll out Red button globally.

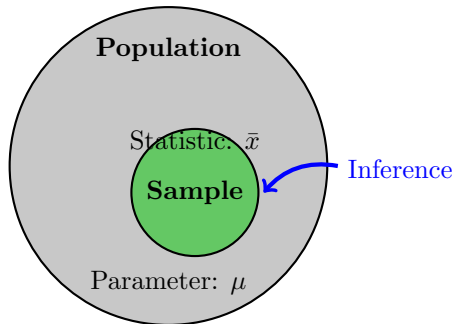
Population vs. Sample

Population (Ω)

- The entire group you want to draw conclusions about.
- *Example:* All 10 million registered voters in a country.
- Measured by **Parameters** (μ, σ).

Sample (S)

- A subset of the population used to collect data.
- *Example:* 1,000 voters surveyed by phone.
- Measured by **Statistics** ($\bar{x}, s$).



Parameters vs. Statistics

A **Parameter** is a fixed (but usually unknown) numerical value describing a population. A **Statistic** is a random variable calculated from sample data.

Measure	Population Parameter	Sample Statistic
Mean	μ	$\bar{x}$
Variance	σ^2	s^2
Std. Deviation	σ	s
Proportion	P or π	$\hat{p}$
Size	N	n

Why Estimate?

Why not just calculate the Population Parameter directly?

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Why not just calculate the Population Parameter directly?

- **Cost:** Surveying 300 million people costs billions.
- **Time:** It would take years to measure everyone; by then, the data is old.
- **Feasibility:** Some testing is destructive.
 - *Example:* To test the safety of a car, you have to crash it. If you test the entire population, you have zero cars left to sell!
- **Accessibility:** We often cannot access the whole population (e.g., all fish in the ocean).

Part 2: Estimation Theory

Estimators and Estimates

Estimator: A rule or formula used to calculate an estimate based on the sample (e.g., the formula $\bar{X} = \frac{\sum X_i}{n}$). This is a Random Variable.

Estimate: The specific numerical value obtained (e.g., $\bar{x} = 5.4$).

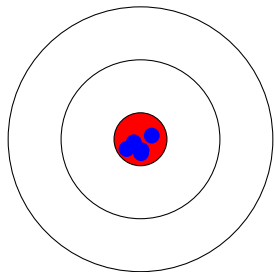
Properties of a Good Estimator:

- ① **Unbiasedness:** On average, the estimator hits the true parameter. $E[\hat{\theta}] = \theta$.
- ② **Efficiency (Low Variance):** The estimates are clustered closely together.
- ③ **Consistency:** As sample size $n \rightarrow \infty$, the estimate converges to the parameter.

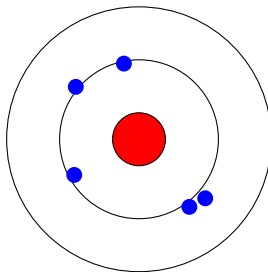
Visualizing Bias and Variance

Think of estimation as throwing darts at a target (the true parameter).

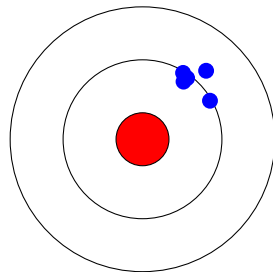
Low Bias, Low Var



Low Bias, High Var



High Bias, Low Var



Proof: Sample Mean is Unbiased

Goal: Prove that $E[\bar{X}] = \mu$.

Assumptions: Let $X_1, X_2, \dots, X_n$ be independent random variables drawn from a population with mean μ .

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Conclusion: Sample Mean ($\bar{X}$) is an unbiased estimator of Population mean (μ).

Sample Variance vs. Population Variance

Population Variance

$$\sigma^2 = \frac{\sum (X_i - \mu)^2}{N}$$

- Denominator is N .
- We know the true center μ .

Sample Variance

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1}$$

- Denominator is $n - 1$.
- We use estimated center $\bar{x}$.

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Degrees of Freedom ($n - 1$): When we calculate $\bar{x}$, we use up 1 unit of information. Only $n - 1$ data points are free to vary. If you know $\bar{x}$ and $n - 1$ values, the last value is fixed.

Proof: Why divide by $(n - 1)$? (Part 1)

Let's define a *biased* variance $S_n^2 = \frac{1}{n} \sum (X_i - \bar{X})^2$. We want to find the expectation $E[S_n^2]$.

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Rearranging for our sample sum of squares:

$$\sum (X_i - \bar{X})^2 = \sum (X_i - \mu)^2 - n(\bar{X} - \mu)^2$$

Proof: Why divide by $(n - 1)$? (Part 2)

Take Expectation of both sides:

$$E \left[\sum (X_i - \bar{X})^2 \right] = E \left[\sum (X_i - \mu)^2 \right] - E \left[n(\bar{X} - \mu)^2 \right]$$

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But, $E \left[\frac{1}{n-1} \sum (X_i - \bar{X})^2 \right] = \frac{1}{n-1}(n - 1)\sigma^2 = \sigma^2$. (Unbiased!)

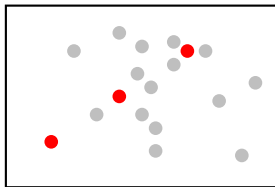
Part 3: Sampling Strategies

Simple Random Sampling

Description: Every member of the population has an equal probability of being selected.

Motivation: Removes systematic bias. Ideally representative.

Use Case: Lottery selection; Random Number Generators, Manufacturing quality control.



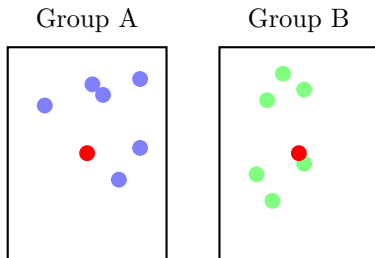
Pure Randomness

Stratified Random Sampling

Description: Divide population into subgroups (strata) based on a characteristic (e.g., Gender, Age), then random sample from each.

Motivation: Ensures minority groups are represented.

Use Case: Election polling (ensure rural/urban mix matches census), market research for different age groups, Popularity of a service among male/female.



Systematic Sampling

Description: Select every k -th individual from a list.

Motivation: Simpler to implement than simple random sampling manually.

Use Case: Quality control on a conveyor belt (test every 100th car part).



Every 3rd item selected

Cluster Sampling

Description: Divide population into clusters (usually geographic). Randomly select **whole clusters**.

Motivation: Cost reduction when population is spread out.

Use Case: Surveying households in a large city (pick specific city blocks and survey everyone there).



Convenience Sampling

Description: Selecting individuals who are easiest to reach.

Motivation: Quick, cheap, easy.

Use Case: Asking your friends/colleagues.

Risk: Highly biased (not representative).

“Hey, can you fill out this survey?”

Part 4: Sampling Distribution & CLT

Sampling Distribution

Definition: The probability distribution of a given statistic (like the mean $\bar{x}$) based on a large number of random samples.

Imagine taking 1000 different samples from a population and calculating the mean for each. If you plot those 1000 means, you get the Sampling Distribution of sample mean.

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Properties of Sampling Distribution of Mean

- **Mean:** $\mu_{\bar{x}} = \mu$ (Unbiased)
- **Variance:** $\sigma_{\bar{x}}^2 = \frac{\sigma^2}{n}$
- **Standard Error:** $SE = \frac{\sigma}{\sqrt{n}}$

Note: As sample size n increases, the spread (SE) decreases!

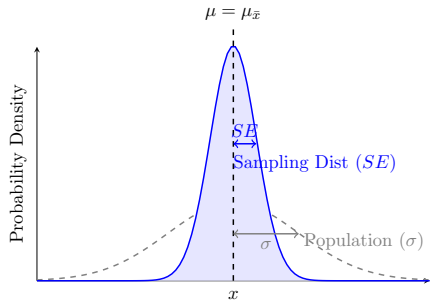
Visualizing Sampling Distribution

Key Concepts:

- **Center ($\mu_{\bar{x}}$):** Stays the same as the Population Mean (μ).
- **Spread (SE):** The sampling distribution is **narrower**.
- **Relationship:**

$$SE = \frac{\sigma}{\sqrt{n}}$$

- As $n \uparrow$, the curve gets thinner (more precision).



Proof: Variance & Standard Error of Sampling Distribution

Objective: Calculate the spread (variance) of the sample mean.

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Property of Variance: $Var(aX) = a^2 Var(X)$. Here $a = \frac{1}{n}$. Hence, $Var(\bar{X}) = \frac{1}{n^2} Var\left(\sum_{i=1}^n X_i\right)$

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Since samples are **independent**, the variance of a sum is the sum of variances:

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Standard Error (SE)

Standard Deviation is the square root of Variance:

$$SE = \sqrt{Var(\bar{X})} = \sqrt{\frac{\sigma^2}{n}} = \frac{\sigma}{\sqrt{n}}$$

Central Limit Theorem (CLT)

Theorem

For a sufficiently large sample size, the sampling distribution of the sample mean will approach a **Normal Distribution**, regardless of the shape of the population distribution.

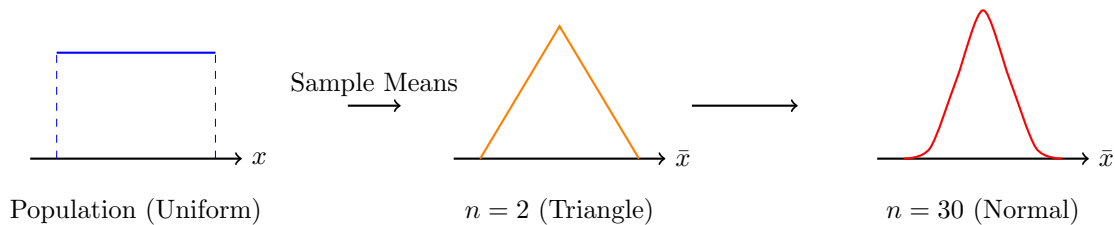
Sampling Distribution $\sim N(\mu, \sigma/\sqrt{n})$ as $n \rightarrow \infty$

Why is this powerful? It allows us to use Normal probability calculations for sampling distribution even if the original data is differently distributed (e.g. Exponential, Uniform, etc.)

How large must n be?

- **Rule of Thumb:** Often cited as $n \geq 30$, but this is not a strict law.
- **Normal Population:** Sampling distribution is Normal for **any** n .
- **Symmetric Population:** CLT works well for small n (e.g., $n > 10$).
- **Highly Skewed Population:** May require larger n (e.g., $n > 50$).

Visualizing CLT



Quiz: Test Your Skills

Q1: Bias

Suppose we estimate the population variance using $S^2 = \frac{\sum (x - \bar{x})^2}{n}$. Which of the following is true?

- A) This estimator is unbiased.
- B) This estimator tends to overestimate the true variance.
- C) This estimator tends to underestimate the true variance.
- D) The bias depends on the population distribution.

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Answer: C) It underestimates. *Reasoning:* The math showed $E[S_n^2] = \frac{n-1}{n}\sigma^2$, which is less than σ^2 .

Q2: Standard Error

If you increase your sample size from $n = 100$ to $n = 400$, what happens to the Standard Error of the mean?

- A) It stays the same.
- B) It is cut in half.
- C) It decreases by a factor of 4.
- D) It doubles.

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Answer: B) *Reasoning:* $SE = \frac{\sigma}{\sqrt{n}}$. If $n_{new} = 4n$, then $\sqrt{4n} = 2\sqrt{n}$. The denominator doubles, so SE is halved.

Q3: CLT Application

A population has a mean of 50 and standard deviation of 10. You take samples of size $n = 4$. Can you assume the sampling distribution of the mean is Normal?

- A) Yes, CLT always applies.
- B) No, the sample size is too small.
- C) Only if the original population is Normal.
- D) Yes, because the mean is large enough.

Q3: CLT Application

A population has a mean of 50 and standard deviation of 10. You take samples of size $n = 4$. Can you assume the sampling distribution of the mean is Normal?

- A) Yes, CLT always applies.
- B) No, the sample size is too small.
- C) Only if the original population is Normal.
- D) Yes, because the mean is large enough.

Answer: C) *Reasoning:* CLT requires large n ($\approx 30+$) if the population is not normal. With $n = 4$, the sampling distribution is only normal if the population itself is normal.

Q4: Sampling Strategy

You want to survey students at a university about campus safety. You stand by the main library entrance and ask people who walk by. What type of sampling is this?

- A) Simple Random Sampling
- B) Cluster Sampling
- C) Convenience Sampling
- D) Stratified Sampling

Q4: Sampling Strategy

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- C) Convenience Sampling
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Answer: C) Convenience Sampling. *Reasoning:* You are prone to bias (e.g., library users might differ from gym users).

Q5: Estimator Properties

Which property ensures that as we collect more data, our estimate gets closer to the true parameter probability 1?

- A) Unbiasedness
- B) Consistency
- C) Efficiency
- D) Normality

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Answer: B) Consistency. *Reasoning:* Unbiasedness is about the “average” attempt. Consistency is about convergence as $n \rightarrow \infty$.

Q6: Degrees of Freedom

Why do we lose a degree of freedom when calculating sample variance?

- A) Because the sample is smaller than the population.
- B) Because we use the sample mean $\bar{x}$ in the calculation, which constrains the data.
- C) To make the calculation easier.
- D) It is a correction for outliers.

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- C) To make the calculation easier.
- D) It is a correction for outliers.

Answer: B) *Reasoning:* Once $\bar{x}$ is determined, if you know $n - 1$ values, the n -th value is mathematically fixed.

Q7: Calculation

Population $\mu = 100, \sigma = 20$. Sample size $n = 25$. What is the probability that the sample mean $\bar{x}$ deviates from 100 by more than 4? (approx 1 std deviation of sampling dist).

- A) 5%
- B) 32%
- C) 68%
- D) 95%

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- A) 5%
- B) 32%
- C) 68%
- D) 95%

Answer: B) *Reasoning:* $SE = 20/\sqrt{25} = 4$. So “4” is exactly 1 SE. The area *within* 1 SE is 68%. The area *outside* is $100 - 68 = 32\%$.

Q8: CLT and Sampling Distribution

Problem: A delivery company has an average delivery time of $\mu = 40$ minutes with a standard deviation of $\sigma = 10$ minutes.

You take a random sample of $n = 25$ deliveries. According to the Central Limit Theorem, what is the approximate probability that the **sample mean** $\bar{x}$ is between 36 and 44 minutes?

- A) 68%
- B) 95%
- C) 99.7%
- D) 100%

Q8: CLT and Sampling Distribution

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- B) 95%
- C) 99.7%
- D) 100%

Answer: B) 95%

Reasoning:

- ① Calculate Standard Error: $SE = \frac{\sigma}{\sqrt{n}} = \frac{10}{\sqrt{25}} = \frac{10}{5} = 2$.
- ② Analyze the Interval: The range $[36, 44]$ is $[40 - 4, 40 + 4]$.
- ③ Convert to Z-scores: Since $SE = 2$, the deviation of 4 is exactly $2 \times SE$.
- ④ Conclusion: The probability of falling within 2 standard deviations ($\mu \pm 2SE$) is approx **95%**.

Summary

- ① **Inference:** Moving from Sample $\rightarrow$ Population.
- ② **Estimators:** We want Unbiased, Consistent, Efficient estimators.
- ③ **Variance:** Divide by $n - 1$ to correct bias.
- ④ **Sampling:** Randomness is key to representative data.
- ⑤ **CLT:** The magic that makes averages Normal, enabling most statistical tests.

Thank You!